Two Pointers for Pair Sum

Topic: CompetitiveProgramming File: c3.pdf

PROBLEM

Given a sorted array, determine whether two distinct elements sum to target T.

APPROACH

Start one pointer at the beginning and one at the end. If their sum is too small, move the left pointer right; if too large, move the right pointer left.

EXAMPLE

A = [1,2,4,6,8], T = 10. The endpoints 1+8=9; move left. Then 2+8=10, so a pair is found.

COMPLEXITY

Time: O(n), since each pointer moves in only one direction. Extra space: O(1).

REQUIREMENT

The array must be sorted. Sorting an unsorted array first costs O(n log n) and may change original indices.